Machine-learning readout of a colorimetric CO_2-sensing ink

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Abstract: A low-cost route to CO_2 monitoring is a colorimetric ink whose colour tracks CO_2, read by miniature reflectance sensors. We characterise one ink over a 77-hour gas-chamber protocol sweeping CO_2 (∼100–1700 ppm), relative humidity (RH) and illumination, recording four red–green–blue–infrared pixels and NDIR and RH references. After validating every channel against these references, we show the blue-to-red ratio B/R responds linearly to CO_2 (sensitivity $\approx -5 \times 10^{-6}$ per ppm), with order-10^3 ppm per %RH humidity cross-sensitivity (loosely constrained) and a lag up to tens of minutes. Humidity is recovered from the ink alone (MAE 0.35 %, $R^2 = 0.993$). For absolute CO_2, dividing each colour channel by the co-measured infrared removes most of the slow drift the channels share: a 12-feature linear model reaches MAE 103 ppm ($R^2 = 0.77$) with no correction, and 50 ppm with one offset per lamp change, beating a hold-the-last-value baseline (54 ppm, R^2 0.71 vs 0.57) on the steady-lamp 34 % of test.
Keywords: colorimetric sensing, CO_2 sensor, optical reflectance, machine learning, regression, sensor drift
SDGs: 3, 9, 11, 13 (see page 6)

I. INTRODUCTION

Carbon dioxide is the standard proxy for indoor air quality and ventilation: occupied rooms drift from the ∼420-ppm background to beyond 1500 ppm, levels with measurable cognitive effects [1]. The reference technology (NDIR) is accurate but bulky and power-hungry. An attractive low-cost alternative is a *colorimetric* sensor: a dye-based ink whose colour depends on the surrounding CO_2 concentration, read out by nothing more than a light source and a photodetector [2]. The film studied here is a prototype from a water-based, CO_2-selective ink family recently developed by the host group [3].

The transduction chain is acid–base chemistry. CO_2 dissolves into the water held by the film and hydrates to carbonic acid, which dissociates and lowers the internal pH,

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-, \qquad (1)$$

shifting the equilibrium of a pH-indicator dye between its basic and protonated forms. The two forms absorb in different spectral bands, so the *hue* of the reflected light, the balance between blue and red, tracks the protonated fraction. The modulations are small, ΔB/R $< 1\,\%$ over the full range, which keeps the reflectance–absorbance relation linear whatever the regime; B/R is then, to first order, linear in that fraction. Two consequences follow. First, the natural observable is a channel *ratio*. Each channel measures illumination times film reflectance, $C_i = I_i \rho_i$, so a lamp change that rescales every I_i by a common factor cancels identically (in the ideal case) in $\mathrm{B/R} = \rho_\mathrm{B}/\rho_\mathrm{R}$, while any single channel is dominated by illumination. B and R make a natural pair: for the indicator dyes used in CO_2-sensing films [2] the two forms absorb at opposite ends of the visible, so ρ_B and ρ_R move in opposite directions. The cancellation is only as exact as the LED's spectral stability under duty changes; Sec. III measures how well it holds. Second, the protonated fraction obeys the Henderson–Hasselbalch relation. The protonated-to-basic ratio is proportional to [H^+], which in a base-buffered film is proportional to the CO_2 concentration [2], so the response is smooth, monotonic and close to linear while that ratio stays small. Equation (1) also exposes the two intrinsic weaknesses. Water activity enters the equilibrium directly, and the film's water content also sets its swelling and the indicator's effective pK_a: humidity must interfere [3]. The second weakness is kinetic. Reaching the equilibrium of Eq. (1) takes gas dissolution and diffusion through the film (the hydration step itself adds only seconds), so the ink is *slow*, with effective time constants of minutes to tens of minutes, dominated by transport.

This slowness set the experimental design: long quasi-static CO_2 triangles (∼11 h per cycle) the film can track; an RH staircase held for hours so cross-talk maps separately; LED-duty steps to expose what the ratios fail to cancel; and a deliberately fast final ∼11 h, a stress test reserved for evaluation.

The goal is a sensor as close to *self-contained* as the ink allows: CO_2 and humidity from its own sixteen numbers, quantifying how far that gets with no reference versus one maintenance offset, for a prototype EMERGE ink. Section III establishes that the ink senses CO_2 quantitatively. Section IV develops the readout: humidity is recovered directly; direct regression (a ladder of models from least squares to recurrent networks under one fixed evaluation protocol) fails against the film's slow drift, and *how* each rung fails is itself informative; the decisive step is that the sensor carries its own drift reference, the infrared channel; one maintenance offset provides the remaining factor of two.

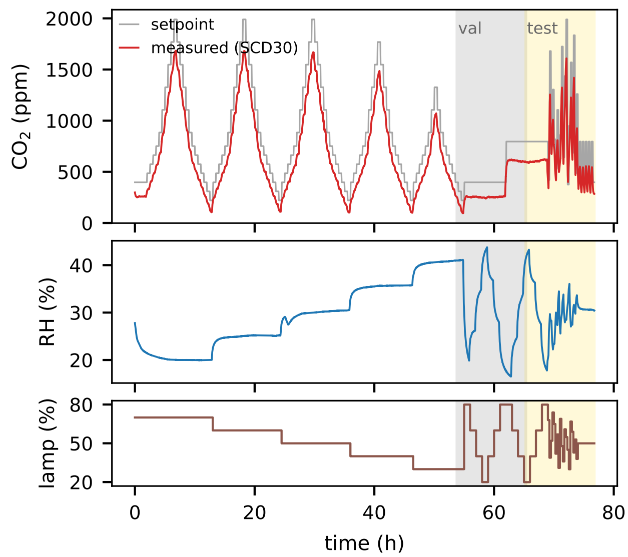

FIG. 1: The 77 h protocol. Top: programmed CO_2 setpoint (grey) and the NDIR reference (red); the chamber tracks the setpoint ($r = 0.97$) with a systematic shortfall at the peaks, one more reason to calibrate against the measured reference. Middle: measured relative humidity. Bottom: LED duty cycle. Shaded bands mark the validation (grey) and test (yellow) periods; everything before them is training data.

II. EXPERIMENTAL SETUP AND DATASET

A. Sensing element and chamber protocol

The ink is deposited as a film and read in reflectance by four identical OnePixel sensors [4] (pixels 0–3), each reporting four channels, red (R), green (G), blue (B) and infrared (IR), at 0.2 Hz under LED illumination whose duty cycle ("lamp") is programmable. Both multiplicities are deliberate: separating hue from illumination needs several colour channels (Sec. IV shows a single one fails), and four identical pixels give dropout redundancy, noise averaging, and a measure of film inhomogeneity that enters the feature set below. The film sits in a sealed chamber fed by three mass-flow-controlled gas lines (dry synthetic air, humidified air, CO_2) [3], with two co-located references: a Sensirion SCD30 NDIR [5] for CO_2 and a Bosch BME280 [6] for RH and temperature.

The 77-hour protocol (Fig. 1): five slow triangular CO_2 sweeps between ∼100 and ∼1700 ppm (∼11 h each), an RH staircase between 17 and 44 %RH, lamp-duty steps between 20 and 80 %, then a deliberately fast final block modulating everything simultaneously. All streams were resampled onto a common 5-s grid (55 258 samples, 16 ink channels).

B. Data validation

Before fitting anything, we identified each logged channel by cross-correlating it with the two references rather than by its column name, and this proved essential. The channel logged as humidity dosing is the one that follows the NDIR CO_2 reference ($r = 0.97$), so we assign every channel by its measured behaviour. The nominal setpoint also sits somewhat above the chamber reading (the SCD30 reaches ∼75 % of nominal, ∼64 % when the humid line runs high), as expected when part of the flow is dilution air. All results below are therefore calibrated against the *measured* references, SCD30 for CO_2 and BME280 for RH, and the true CO_2 range is the ∼100–1700 ppm of Fig. 1.

C. Learning task and splits

The supervised task is joint regression of the measured pair (RH, CO_2) from ink reflectance only: the 16 raw channels plus 56 derived features (per-pixel chromaticity ratios such as B/R and B/(R+G+B), and cross-pixel means and standard deviations). Reference channels, setpoints and lamp duty are excluded from the inputs: ratios cancel illumination by construction, and a control adding the lamp duty as an input changes ridge-72 by ∼3 % while degrading the IR-referenced features. Only the *timing* of lamp changes is used, to schedule that offset.

The split is *chronological*, 70/15/15 train/validation/test with 4-min guard gaps (Fig. 1), and this choice matters more than any model choice. Adjacent 5-s samples are nearly identical, since the chamber and the film evolve over minutes, so a *random* split would put near-twins of the test points inside the training set and leakage would look like skill. Every result below is chronological. The price is facing real distribution shift, because here the chronological split is also a *regime* split. The protocol operates in three modes: slow CO_2 triangles with an RH staircase (train), nearly constant CO_2 under lamp and RH changes (validation), and fast simultaneous modulation with a quieter tail (test). Generalising therefore means transferring across operating modes as well as across time. The test block moves CO_2 in 15-min steps, faster than the film can follow (Sec. III), its range narrows to 283–1609 ppm, and the film has aged for days. Validation, with almost no CO_2 variance, can rank RH models but barely CO_2 ones. This mirrors a key deployment difficulty: regime shift without retraining. As floors, predicting the training mean gives test MAE 3.88 %RH and 197 ppm CO_2; a model has to beat *these*, not zero.

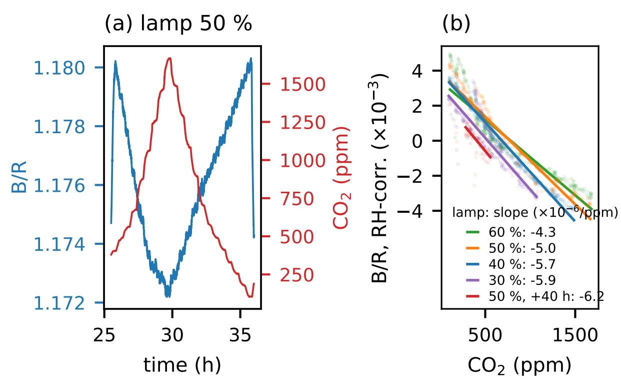

FIG. 2: Colorimetric response. (a) Pixel-averaged B/R (blue, left axis) and reference CO_2 (red, right axis) during the 50 % lamp segment: the ratio mirrors the full sweep; the spikes at the edges are lamp transitions. (b) RH-corrected, mean-centred B/R versus CO_2 for the lamp-stable segments with the joint fits of Eq. (2); the 50 % lamp condition revisited 40 h later (red) gives a slope of the same sign and magnitude.

III. COLORIMETRIC CHARACTERISATION

Within any segment of constant lamp duty (excluding the first settling hour), the pixel-averaged blue-to-red ratio B/R follows the CO_2 sweeps closely once the humidity contribution is accounted for. We fit, per segment, the linear model

$$\mathrm{B/R} = \alpha\,[\mathrm{CO_2}] + \beta\,\mathrm{RH} + \gamma, \tag{2}$$

and report α together with the partial correlation between B/R and CO_2 at fixed RH (Table I, Fig. 2). Three features stand out. First, within the scatter the response shows no detectable curvature over the full 100–1700 ppm range, with sensitivity $\alpha \approx -4$ to -6×10^{-6} ppm^{-1} and partial correlations reaching -0.99, consistent with the dilute limit of the indicator equilibrium [2]. The sign is the expected one: more CO_2 acidifies the film, converting the blue-coloured basic form of the dye (absorbing in the red) into its protonated form (absorbing in the blue), so blue reflectance falls while red recovers and B/R drops. Second, the response *persists*: the 50 % lamp condition re-measured 40 h later yields a slope of the same sign and magnitude (-6.2 vs -5.0×10^{-6}, $\sim$25 %). Lamp and elapsed time are confounded, so the monotonic trend of α down Table I need not be an illumination effect; slow gain drift of the ageing film is the simpler reading, consistent with Sec. IV. Third, humidity is the dominant interferent: a global fit puts the RH term near 10^3 ppm of apparent CO_2 per %RH, although the staircase protocol, which holds RH nearly constant inside lamp-stable segments, leaves β only loosely identified (its per-segment value varies in magnitude and even sign); pinning it down needs a protocol that sweeps RH and CO_2 jointly. Any quantitative use of the ink therefore needs to know RH at the same time, which, as shown next, the ink itself provides.

The response is not immediate: cross-correlating RH-corrected B/R with the NDIR gives a $\sim$3-min apparent lag in slow sweeps but up to $\sim$44 min in the fast block. The film acts as a low-pass filter whose apparent lag grows with excitation speed; fluctuations faster than it are attenuated, not tracked.

IV. PREDICTING RH AND CO_2 FROM REFLECTANCE

A. Models and evaluation

Table II reports every model on identical inputs, splits and metrics, with train next to test error so the *generalisation gap* is visible per family; Fig. 5 (appendix) shows every fit over the full run. Hyperparameters of the snapshot models (one 5-s sample in, no history) were tuned on validation only; it carries almost no CO_2 variance (Sec. II.C), so the LSTM configuration is a representative architecture.

B. Humidity

Humidity turns out to be the easy half. Against the 3.88 %RH floor, a ridge tuned for RH (8-h high-pass window, selected on validation) reaches **MAE 0.35 %RH,** $R^2 = 0.993$ on the test block (Fig. 3a). This is physically expected, since water activity enters Eq. (1) directly. An ablation shows where the signal lives: the red channel alone fails below the floor (4.4 %RH, one band only measures absorbance times illumination), the four colours of a single pixel already recover 0.60 %RH, and the sixteen raw channels together do *worse* than four (1.04). The information is spectral, and the ink itself measures its own dominant interferent.

TABLE I: Per-segment joint fits of Eq. (2) in lamp-stable windows (first settling hour excluded). r_p is the partial correlation of B/R with CO_2 at fixed RH. The initial 70 % lamp sweep (1–13 h) is reported but excluded from the summary: it is the only segment at that lamp level and the only one with a fresh, still-equilibrating film, so the two effects are confounded ($\alpha = -7.5 \times 10^{-6}$, $r_p = -0.68$).

Lamp (%)	t (h)	CO_2 (ppm)	α (10^{-6}/ppm)	r_p
60	14–24.5	108–1680	−4.3	−0.77
50	25.5–36	102–1668	−5.0	−0.95
40	37–46.5	99–1486	−5.7	−0.99
30	47.5–55	93–1070	−5.9	−0.91
50	75–76.7	282–556	−6.2	−0.83

TABLE II: The model ladder, curated to one representative per lesson (Fig. 5 in the appendix shows every fit). Train/test MAE for both targets (RH in %, CO_2 in ppm) and test Pearson r for CO_2; $R^2_{\mathrm{te}} < 0$ for all CO_2 entries except the IR-referenced ridge ($R^2 = 0.77$) and the bottom block ($R^2 = 0.57/0.65/0.71/0.68$). N_p = trainable parameters ("—" = nonparametric). Floors (predict the train mean): 3.88 %RH, 197 ppm. Bottom block: one 30-min offset per lamp change, evaluated on lamp-stable test windows (34 % of test); LSTM r/offset-MAE: five-seed means (other cells one archived run); MLP: three-seed means.

		RH MAE		CO_2 MAE			
Model	N_p	tr	te	tr	te	r_{te}	What it teaches
Ridge, 72 features	146	0.17	0.88	119	403	0.76	physics-guided features help
Ridge + high-pass 4 h	578	0.04	0.77	44	297	0.43	drift lives at low frequency
Ridge, 12 (R,G,B)/IR	26	0.27	1.74	179	**103**	**0.90**	IR reference cancels drift
MLP 128×64, reg.	18k	0.23	1.34	26	521	0.70	regularisation alone did not fix drift
LSTM, 6-h coarse	21k	0.14	0.69	19	654	0.81	long context: shape, not scale
Hold calibration value	0	—	—	—	54	0.79	the bar recalibration sets
B/R + RH + recal.	3	—	—	—	54	0.82	ties the bar, tracks better
(R,G,B)/IR + offset	26	—	—	—	**50**	**0.91**	the deployable readout
LSTM 6-h + offset	21k	—	—	—	48	0.82	regime-sensitive experiment

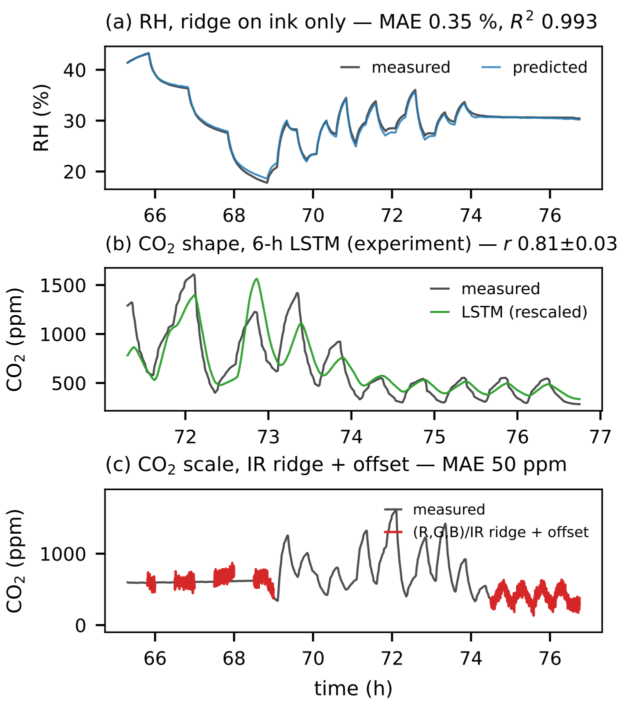

FIG. 3: Test-block predictions. (a) RH from ink reflectance alone. (b) The 6-h-context LSTM tracks the CO_2 oscillations ($r = 0.81 \pm 0.03$ over seeds; best of five shown, drawn after its 6-h warm-up) but loses the absolute scale; for display its output is affinely rescaled, which is precisely the calibration a deployment must supply. (c) The same LSTM with one 30-min offset per lamp change (red, drawn where evaluated) recovers absolute CO_2 in the lamp-stable windows. Fluctuations faster than the film's ~44-min apparent lag are smoothed by the ink itself.

C. CO_2 by direct regression

The same recipes collapse on CO_2: every plain learner ends below the 197-ppm floor, killed by the slow baseline drift of the ageing film. The failures are systematic and each teaches something (Table II). Least squares on the 16 raw channels simply memorises the training period: 87 ppm on train, 1549 on test. Engineered ratios repair part of that (403 ppm) and causal high-pass features a bit more (297 ppm at a 4-h window), but the CO_2 signal rides a baseline whose drift overlaps the protocol cycles, so no static feature set removes it. More capacity makes things worse: growing an unregularised MLP from 4.8k to 2.2M parameters drives training error down while test error climbs from 671 to 1030 ppm, and proper regularisation does not rescue it (521 ± 67 ppm, $r = 0.70$, three seeds). Sequence models come closest: an LSTM reading six hours of 60-s-averaged context tracks the shape of the oscillations better than anything else ($r = 0.81 \pm 0.03$ over five seeds, Fig. 3b) yet still posts MAE ≈650 ppm, because no context length we could test (up to 8 h) supplies the absolute anchor. The pattern repeats in every family, linear to recurrent: none infers the drift from the visible channels; added capacity only memorises it.

D. CO_2 with an infrared reference

The way out is not a bigger model but a better ratio. The dye barely touches the *infrared* channel, so IR shares the slow ageing of emitter, detector and film with the colour channels while staying blind to the gas, and the division cancels what they share: from train to test, B/IR shifts by only 0.26 train-σ, against 1.09 for B/R and 0.69 for raw B. A ridge on the twelve (R,G,B)/IR ratios reaches **MAE 103 ppm,** $R^2 = 0.77$, $r = 0.90$ **with no correction at all**, the only model that beats the constant floor unaided. Its test error is even *lower* than its training error (103 vs 179 ppm), because the test block spans a narrower range; relative to each block's spread both errors are the same $\approx 0.4\sigma$. What the ratio does not cancel are the lamp steps (IR barely follows the duty cycle), and the remaining error is made of exactly that. The fitted weights mostly sit on the red-channel ratios with opposing signs across pixels, which points to between-pixel common-mode rejection (collinear coefficients are not individually meaningful), and tiny MLPs only lose: the relation is essentially linear.

E. CO_2 with one offset per illumination change

After an illumination change, nothing in the ink signal says where the absolute level sits; the standard remedy is one offset, measured against a reference during the first 30 min of each lamp segment (those samples are excluded, leaving 34 % of the test block for evaluation, 283–623 ppm). The bar this sets is strict: just holding the calibration value already gives 54 ppm ($R^2 = 0.57$). The corrected ridge (122 ppm) and MLP (79, one seed) fall short, and the three-parameter Eq. (2) ties it while tracking better ($R^2 = 0.65$), with measured or ink-derived RH alike. The IR-referenced ridge is the one that beats it: **50 ppm,** $R^2 = 0.71$**,** $r = 0.91$ (Fig. 3c), twelve features, selected by validation (Fig. 4). The six-hour LSTM matches it numerically (47.7 ± 1.2 ppm over five seeds), but only while training matches the test regime; in the single refit run of Sec. IV.F it collapses, so we report it as an *experiment* on temporal context. Without the offset, Eq. (2) fails like everything else (171 ppm): the residual drift can only be removed by measuring it.

Three caveats bound these numbers. The offsets are computed against the co-located NDIR reference, so a deployment needs a reference channel or a reference-free scheme (e.g. baseline autocalibration [7]). All results come from one film in one 77-h run. The single-ratio B/R noise after 60-s averaging, $\sim 1.1 \times 10^{-4}$ ($\approx$22 ppm at the measured sensitivity), is an indicative resolution scale; in practice drift dominates the error budget. Chamber temperature (27.3–31.2 °C) correlates weakly with the residuals ($r = 0.26$), so it is not the dominant term.

These limits also say what the ink *is* for. As a ppm meter it needs slow environments and an occasional anchor; as an *indicator*, neither: with no correction at all, the IR-referenced readout reaches AUC 0.98 at 800 ppm and 0.99 at 1000 ppm (accuracies 92 % and 96 % against majority-class baselines of 79 % and 88 %). That is the indoor-ventilation use the host group's monitoring platform targets [3, 4]: a low-cost element asking only whether the level is above a threshold.

F. Adding the validation block to training

As a last experiment we refit on train plus validation, every choice kept fixed, adding lamp and RH variation at constant CO_2 to the training data. Every snapshot model clearly improves: ridge+HP 297→137 ppm, MLP 521→155, the IR ridge 103→86 without correction and 50→45 ppm with the offset (one configuration). The recurrent model instead collapses in our single refit run (r $0.81 \pm 0.03 \rightarrow 0.05$, one seed): the added block apparently teaches it a wrong temporal regime. The binding constraint, then, is the data itself: one film, one run, five similar sweeps, one confounder block.

V. CONCLUSIONS

- The ink is a genuine CO_2 sensor: in lamp-stable conditions B/R shows no detectable deviation from linearity over 100–1700 ppm ($\alpha \approx -5 \times 10^{-6}$ ppm^{-1}, partial correlations to -0.99); the slope keeps sign and magnitude after 40 h ($\sim$25 % gain drift, absorbed by the IR reference).

- Humidity is both the dominant interferent (order 10^3 ppm per %RH, loosely identified) and accurately recoverable from the ink itself on this film (MAE 0.35 %RH, $R^2 = 0.993$).

- Absolute CO_2 readout is limited by slow drift rather than model capacity; most of that drift is channel-common and cancelled by the infrared reference: a 12-feature (R,G,B)/IR linear model reaches 103 ppm ($R^2 = 0.77$) without correction and 50 ppm with one 30-min offset per illumination change; all results are one-film, reference-anchored.

- A central limitation is the *speed* of the CO_2 changes: the apparent lag grows from $\sim$3 to $\sim$44 min under fast modulation. Fluctuations faster than this are physically unrecoverable, a floor on test error distinct from model failure (Appendix); this sets the application envelope (slowly varying air); next: drift-stable references, an orthogonal RH$\times CO_2$ sweep, a multi-film repeat. In threshold mode the absolute-anchor requirement disappears (AUC 0.98 at 800 ppm with no correction): the ventilation-indicator regime.

- Methodologically, validating every channel against independent references before fitting proved decisive: it set the calibration targets the rest of the work relies on.

Acknowledgments

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Lectura per aprenentatge automàtic d'una tinta colorimètrica sensible al CO_2

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Resum: Una via de baix cost per monitorar el CO_2 és una tinta colorimètrica el color de la qual canvia amb la concentració de CO_2, llegida per sensors de reflectància miniaturitzats. Caracteritzem una tinta d'aquest tipus dins d'una cambra de mescla de gasos durant un protocol de 77 hores que escombra el CO_2 ($\sim$100–1700 ppm), la humitat relativa (HR) i la il·luminació, enregistrant quatre píxels de reflectància vermell–verd–blau–infraroig juntament amb una referència NDIR (CO_2) i una d'HR. Després de validar el conjunt de dades amb les referències mesurades, mostrem que la raó blau/vermell respon linealment al CO_2 amb sensibilitat $\approx -5 \times 10^{-6}\,\mathrm{ppm}^{-1}$, una sensibilitat creuada a l'HR de l'ordre de 10^3 ppm per %HR, i un retard de resposta de minuts a desenes de minuts. Comparem models de calibratge de capacitat creixent, des dels mínims quadrats fins a xarxes neuronals recurrents. La humitat es recupera només amb la tinta (MAE 0,35 %, $R^2 = 0{,}993$); per al CO_2 absolut, referenciar cada canal de color a l'infraroig co-mesurat cancel·la la major part de la deriva òptica lenta: un model lineal de 12 característiques assoleix MAE 103 ppm ($R^2 = 0{,}77$) sense cap correcció, i 50 ppm amb un ajust puntual per canvi d'il·luminació, sobre el 34 % del bloc de test estable en il·luminació.
Paraules clau: detecció colorimètrica, sensor de CO_2, reflectància òptica, aprenentatge automàtic, regressió, deriva de sensors
ODSs: 3, 9, 11, 13

Objectius de Desenvolupament Sostenible (ODSs o SDGs)

1. Fi de la pobresa		10. Reducció de les desigualtats	
2. Fam zero		11. Ciutats i comunitats sostenibles	X
3. Salut i benestar	X	12. Consum i producció responsables	
4. Educació de qualitat		13. Acció climàtica	X
5. Igualtat de gènere		14. Vida submarina	
6. Aigua neta i sanejament		15. Vida terrestre	
7. Energia neta i sostenible		16. Pau, justícia i institucions sòlides	
8. Treball digne i creixement econòmic		17. Aliança pels objectius	
9. Indústria, innovació, infraestructures	X		

El contingut d'aquest TFG es relaciona amb l'ODS 3 (Salut i benestar), fita 3.9, ja que el monitoratge de la qualitat de l'aire interior mitjançant sensors de CO_2 de baix cost contribueix a reduir l'exposició de les persones a ambients mal ventilats. Es relaciona també amb l'ODS 9 (Indústria, innovació, infraestructures), fita 9.5, perquè desenvolupa i caracteritza una tecnologia de sensors innovadora i de baix consum; amb l'ODS 11 (Ciutats i comunitats sostenibles), fita 11.6, perquè un monitoratge ambiental assequible i ubic és una eina per a edificis i ciutats més saludables; i amb l'ODS 13 (Acció climàtica), fita 13.3, ja que millora la capacitat de mesura del CO_2, el principal gas amb efecte d'hivernacle d'origen antropogènic.

Appendix A: SUPPLEMENTARY MATERIAL

Training details (Table III)**.** The ridge penalty ($\in \{10, 10^2, 10^3\}$) and the high-pass window were selected on the validation block only. The LSTMs use a single recurrent layer of 64 units trained with Adam (learning rate 10^{-3}, 25–40 epochs, batch 512) on standardised inputs and targets; training windows are strided by 30 s, test windows by 5 s. The coarse-context variant first applies a 60-s moving average to the 16 channels, resamples at one point per minute, and reads windows of 360 such points (6 h); a 12-h window is not measurable because the test block is only 11.5 h long. Window-normalised LSTMs (each input window z-scored independently, 30/60-min windows) end at CO_2 MAE 386/299 ppm with $r = 0.12/0.10$. All models receive identical chronological splits with 4-min guard gaps; stochastic models (MLPs, LSTMs) use a fixed random seed and each configuration is a single run, except the coarse-context LSTM and its recalibrated variant (five seeds: shape r 0.76–0.85; corrected MAE 46.7–49.9 ppm) and the regularised MLP (three seeds). All rolling windows are trailing (causal).

Recalibration policy. The 30-min window after each lamp change is used to fit a single additive offset per model and segment; those samples are excluded from all reported metrics, and windows shorter than 30 min contribute no evaluated samples. Evaluated coverage is 34 % of the test period (3.9 h), spanning 283–623 ppm. Reducing the calibration window to 10 min degrades the physical model to MAE ∼100 ppm; extending it to 60 min shrinks the evaluated windows and also degrades the result.

What else was tried. (0) Raw-window LSTMs (5–30 min of 5-s history) reach only 574–582 ppm: the relevant history exceeds any affordable raw window, which motivated the coarse-context design. (i) A derivative-predicting RNN integrated from each anchor (dead reckoning, 10-seed ensemble) won validation decisively but lost on test (61 ppm): a derivative bias of only 0.025 $\mathrm{ppm\,s^{-1}}$ is invisible in dynamics-rich validation yet fatal in flat test windows: regime mismatch can silently invert rankings. (ii) Kinetic features (d/dt, d^2/dt^2 of ratios and channels at 1–60 min scales) all lost to their static controls, and lead-compensating the film's lag was rejected at every τ from 3 to 90 min: differentiation amplifies the RH signal; lag compensation is not the lever. (iii) A PatchTST-style transformer (105k params, val-selected, 3 seeds) lost to the LSTM on every metric (85 vs 48 ppm corrected; shape $r \approx -0.4$). (iv) Adversarial RH-decoupling via gradient reversal only hides RH from the online head: a fresh probe still recovers RH at $R^2 \approx 0.93$, and forcing real invariance collapses CO_2 in lockstep: the two signals share the protonation pathway and cannot be separated in this ink.

Train+validation refit. Full numbers of the Sec. IV.F refit: ridge-72 403→250 (R^2 still < 0); ridge+HP 297→137 ($R^2 = 0.48$); MLP 521→155 ($R^2 = 0.33$); IR ridge 103→86 ($R^2 = 0.80$, $r = 0.92$) and, with the offset, 50→45 ppm ($R^2 = 0.76$); B/R+RH+offset 54→53; LSTM 6-h shape r 0.81→0.05 and 48→93 ppm with offset. The validation block teaches confounder robustness to snapshot models and a wrong temporal regime to recurrent ones.

Reproducibility. The full pipeline (feature construction, every model, figure and verification scripts) is a documented set of Python scripts, available on request; total training time is under five minutes on CPU for all linear models plus ∼10 min on one GPU for the LSTM variants.

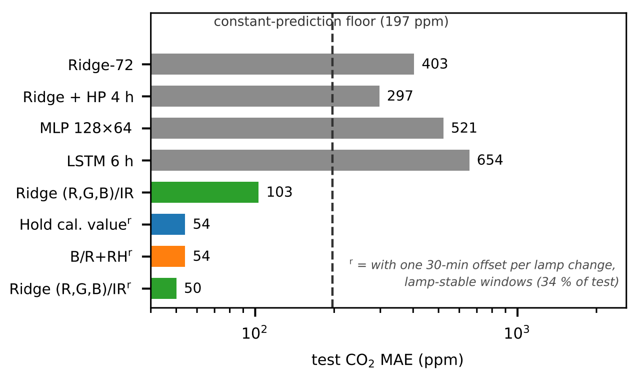

FIG. 4: Summary: test CO_2 MAE of the main models (log scale). Direct regression never beats the constant-prediction floor; the IR-referenced ridge beats it with no correction (103 ppm); with one 30-min offset per lamp change, it and the six-hour LSTM tie at 50/48 ppm.

TABLE III: Configurations behind Table II. All models: standardised inputs, chronological splits; ridge/MLP hyperparameters chosen on validation.

Model	Inputs	Key settings
Ridge	72 feat.	$\alpha \in \{10, 10^2, 10^3\}$ on val
Ridge + HP	72 + 216 high-pass	HP windows 1/2/4/8 h, + 5/15-min smooth
MLP	72 feat.	ReLU 128×64, drop 0.2, wd 10^{-4}, early stop
LSTM (coarse)	16 raw, 60-s MA, 1/min	W=360 (6 h), hidden 64, 40 epochs
B/R + RH	pixel-mean B/R, RH	least squares on train; 30-min offset

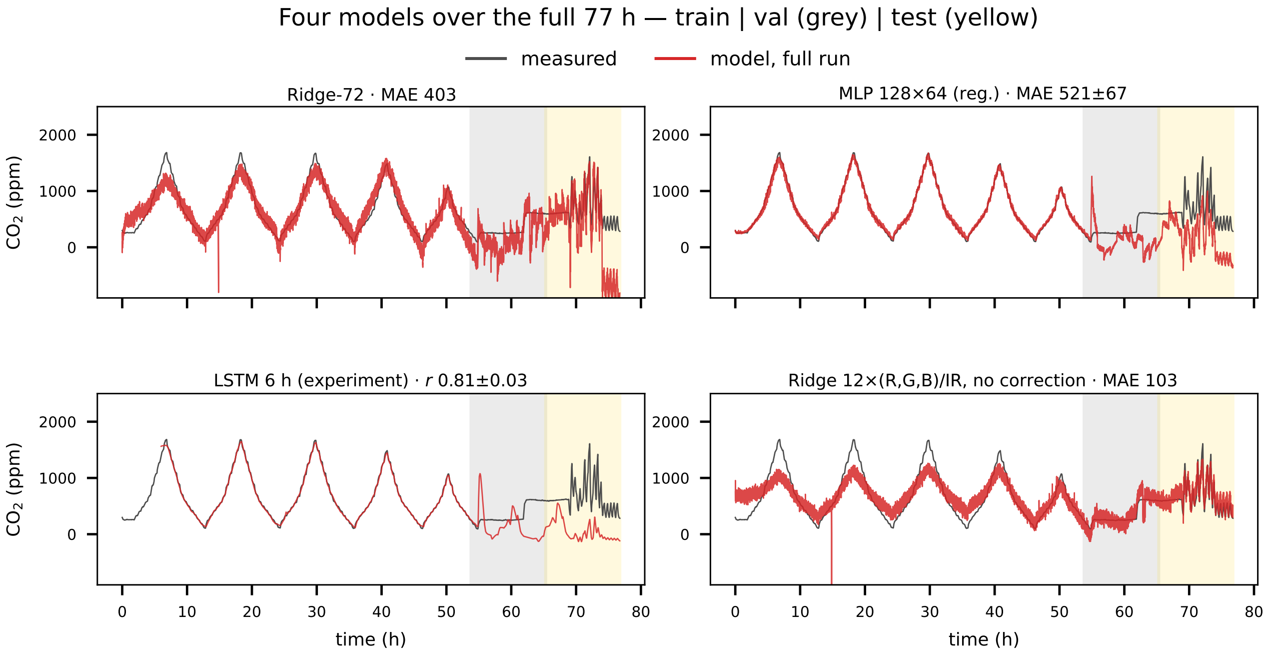

FIG. 5: Four models over the full 77 h run, every one fitted on the training block only: prediction (red, clipped to the axis range) against the NDIR reference (grey); validation (grey band) and test (yellow band) marked. Each model fits the training period; the bands reveal who generalises. The large MLP and Ridge-72 lose the scale to drift; the LSTM keeps the shape; only the uncorrected IR-referenced ridge stays on the measured trace throughout. Quoted MAEs are the test values of Table II.